Key Service System Integration

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ETT SMART | **ETT SMART**

Hotell & Lägenheter

CNS Application
Tackjärnsvägen 13 • SE-168 68 Bromma • Mobile: +46 (0) 705932899 • Mail: pkp@cnsapp.se • SkypeID: pk_koster • Org.nr 556530-5629

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Key Service System Integration

Background

The system integrator (KSSI) has been designed to retrieve today's guest arrivals from any Property Management System (PMS) ,that supports a booking retrieval function on Internet. Furthermore the KSSI communicates with a Winkey Webservice from dormakaba and based on the information retrieved from the PMS, the Webservice generates an access code.

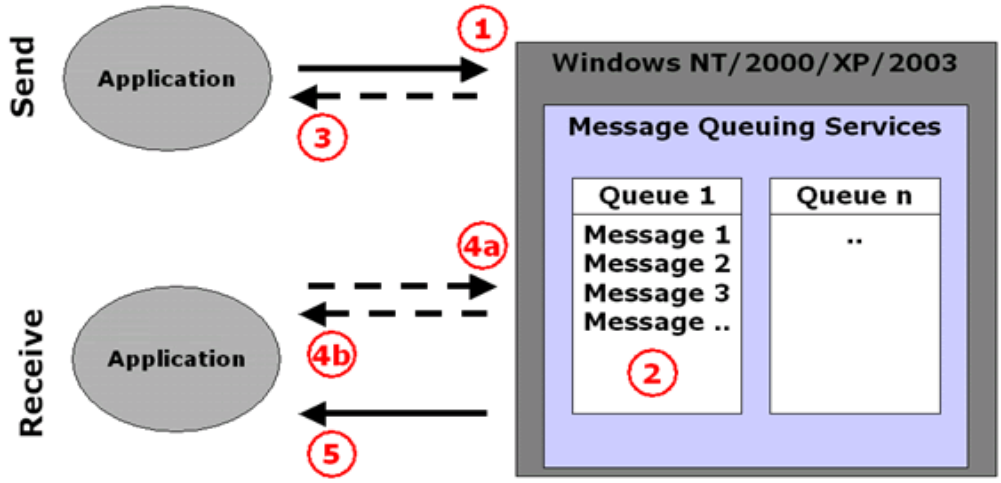
PMS from HappyBooking is the current PMS enabled (2020-09-10) on the production environment.

MSMQ overview

Microsoft® Message Queue Server (MSMQ) makes it easy for application developers to communicate with application programs quickly, reliably, and asynchronously by sending and receiving messages.
We have a lot of opportunities with MSMQ, but we have chosen to work with private queues only.
We have used 4 properties, when interacting with MSMQ:

- **Label** – property informative text, like 'Integrator_ + <the companyID of the property>'
- **AppSpecific** - property to special case handling for certain messages, like:
  1 for 'Generate messages for mobile phone and email'.
  2 for 'Generate Journal message'
  9 for 'Database maintenance (processed only once per day)'
  10 for 'Generate Alert email message and XML document to the hosting provider.

- **Body** – property message content, like passing a whole XML document.
- **Delivery** – property for various delivery options. See below.
- (
- *-- MQMSG_DELIVERY_RECOVERABLE - In every hop along its route, the message is forwarded to the next hop
- *-- or stored locally in a backup file until delivered. This guarantees delivery even in the case of a machine crash.
- *-- MQMSG_DELIVERY_EXPRESS - The default. The message stays in memory until it can be delivered.
- *-- (In-memory message store and forward.)
- *--
- *-- When the message's delivery mechanism is set to MQMSG_DELIVERY_EXPRESS, the message has faster
- *-- throughput.
- *-- When set to MQMSG_DELIVERY_RECOVERABLE, throughput may be slower. However, MSMQ guarantees that
- *-- the message will be delivered, even if a computer crashes while the message is en-route to the queue.
-)



(1) Application sends message to MSMQ
(2) MSMQ stores message in a queue
(3) MSMQ confirms that message is stored
(4a) Application requests a stored message or
(4b) MSMQ notifies application about new messages (Trigger)
(5) Application receives message

„Command" Message

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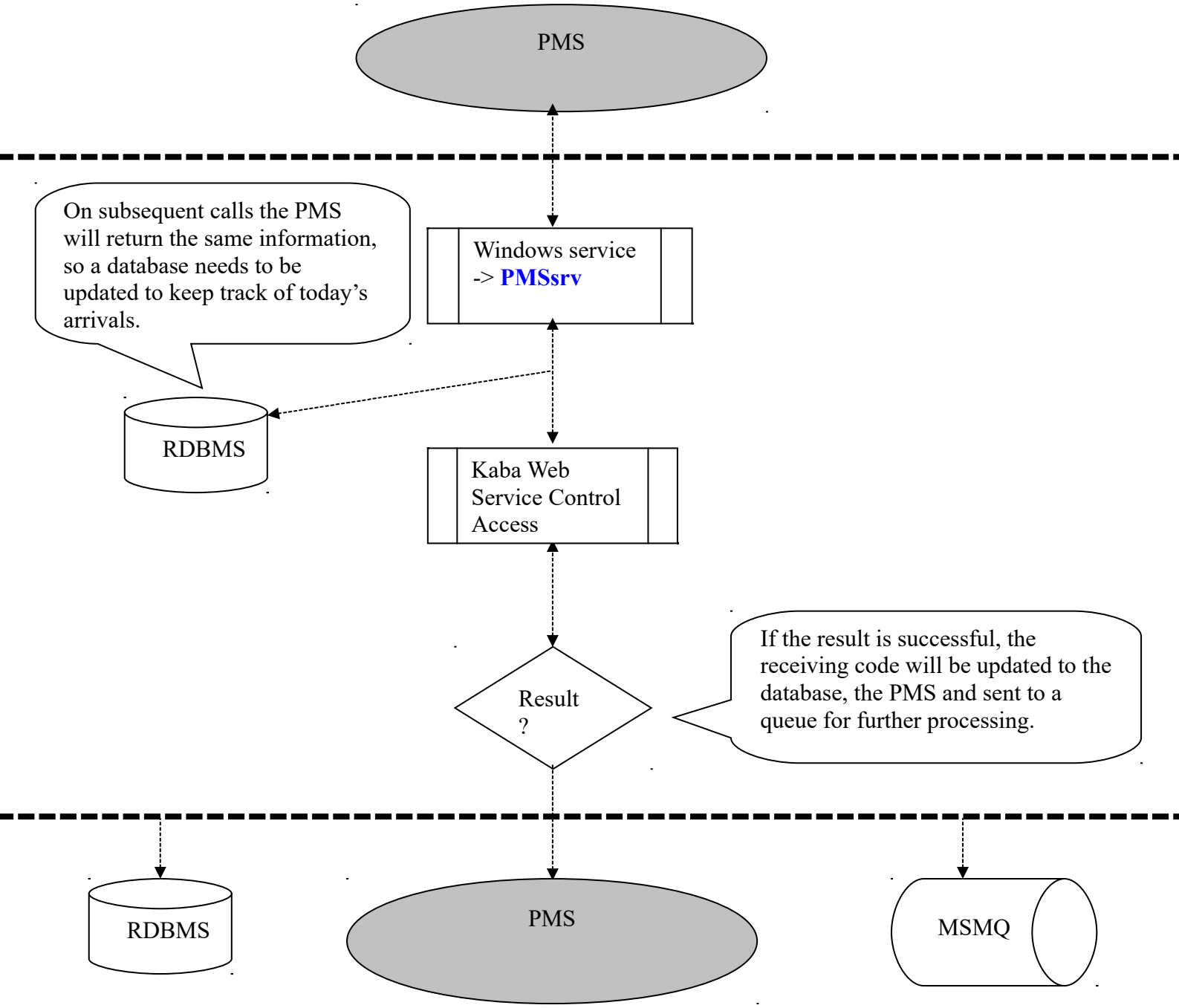
KSSI HTTP handler

The communication between PMS and the KSSI depends on protocol support from the PMS. For PMS HappyBooking a HTTP connection with base64 encoding for authorization is used and information from the PMS will be returned in JavaScript Object Notation (JSON) format.



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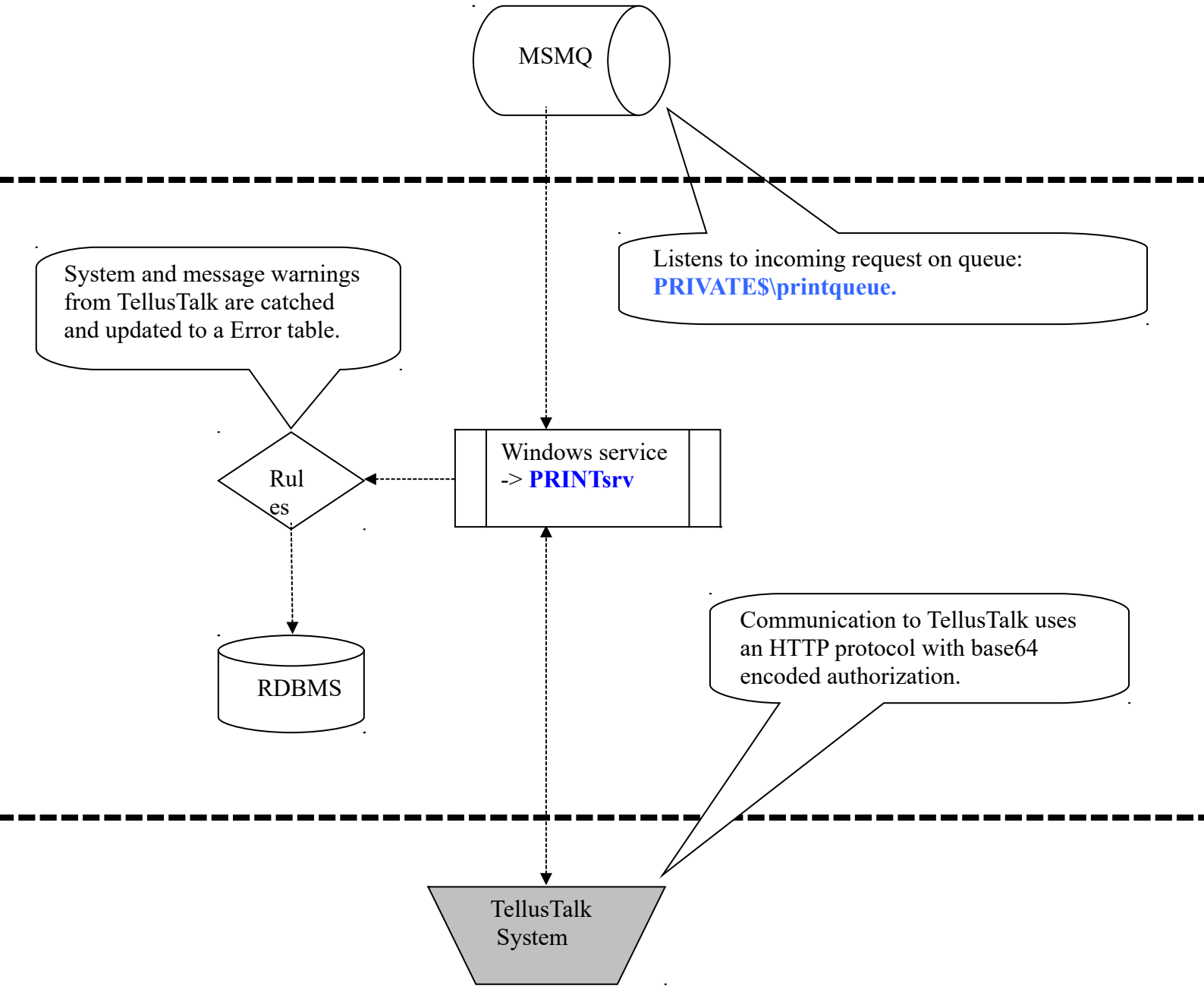
KSSI Mailing process

The KSSI Mailing process will listen to incoming request on queue: PRIVATE$\printqueue and as soon as a message hits the queue, processing will start. All emailing and SMS uses components from TelluysTalk for the actual mailing. The text used for the mailing relies heavily on configuration file and script files.



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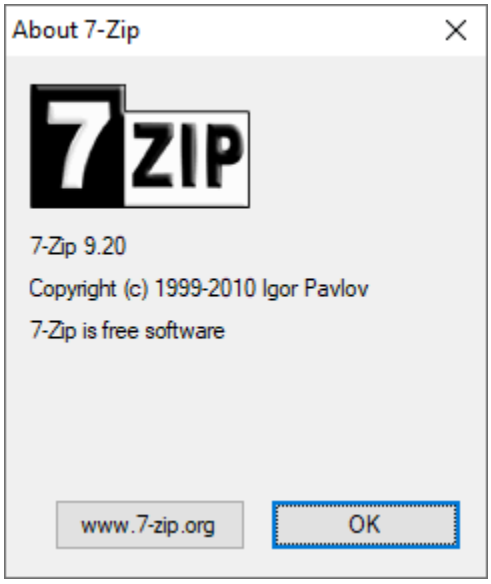
Zip process

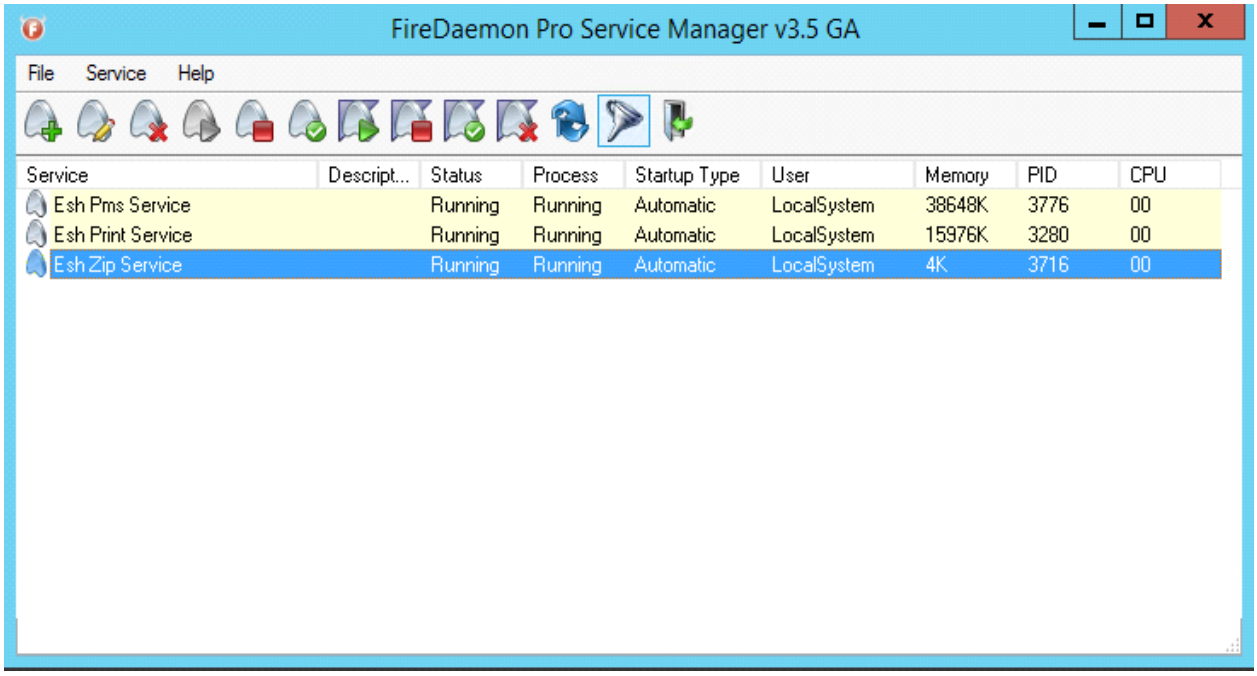
Once per day (23:50) a Zip process is executed moving all files into a zip directory. 7-Zip has been the selected tool for this purpose. See screenshot below.



FireDaemon Service Manager

From FireDaemon users guide:
FireDaemon Pro provides an intelligent interface or "wrapper" between the commands issued by the Service Control Manager (SCM) and your application. It does not modify your application in any way. FireDaemon Pro monitors your application for you plus provides a variety of additional features that allow you to fine tune how your application starts, what to do if and when it stops and how to interact with the rest of the system and environment.

This tool was chosen due to superior functionality compared to Windows service manager.



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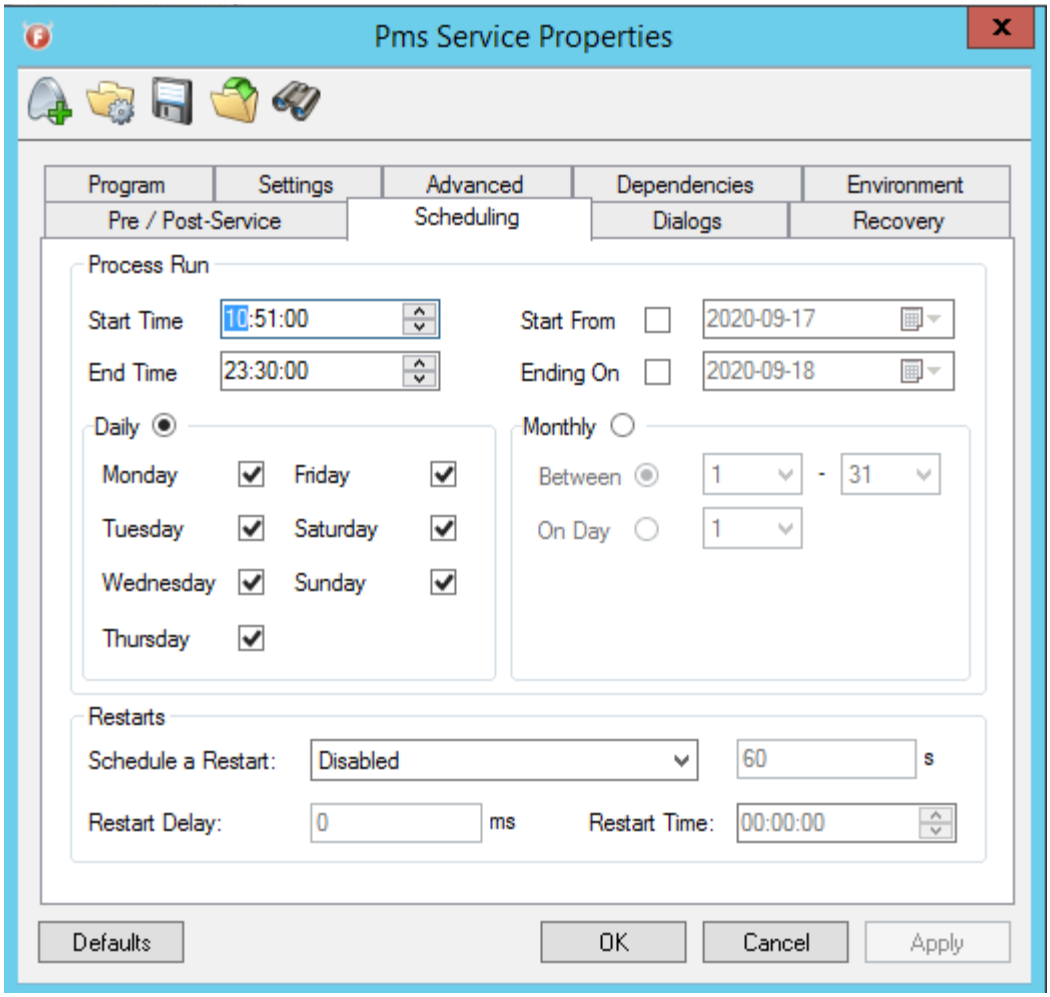
FireDaemon properties implemented

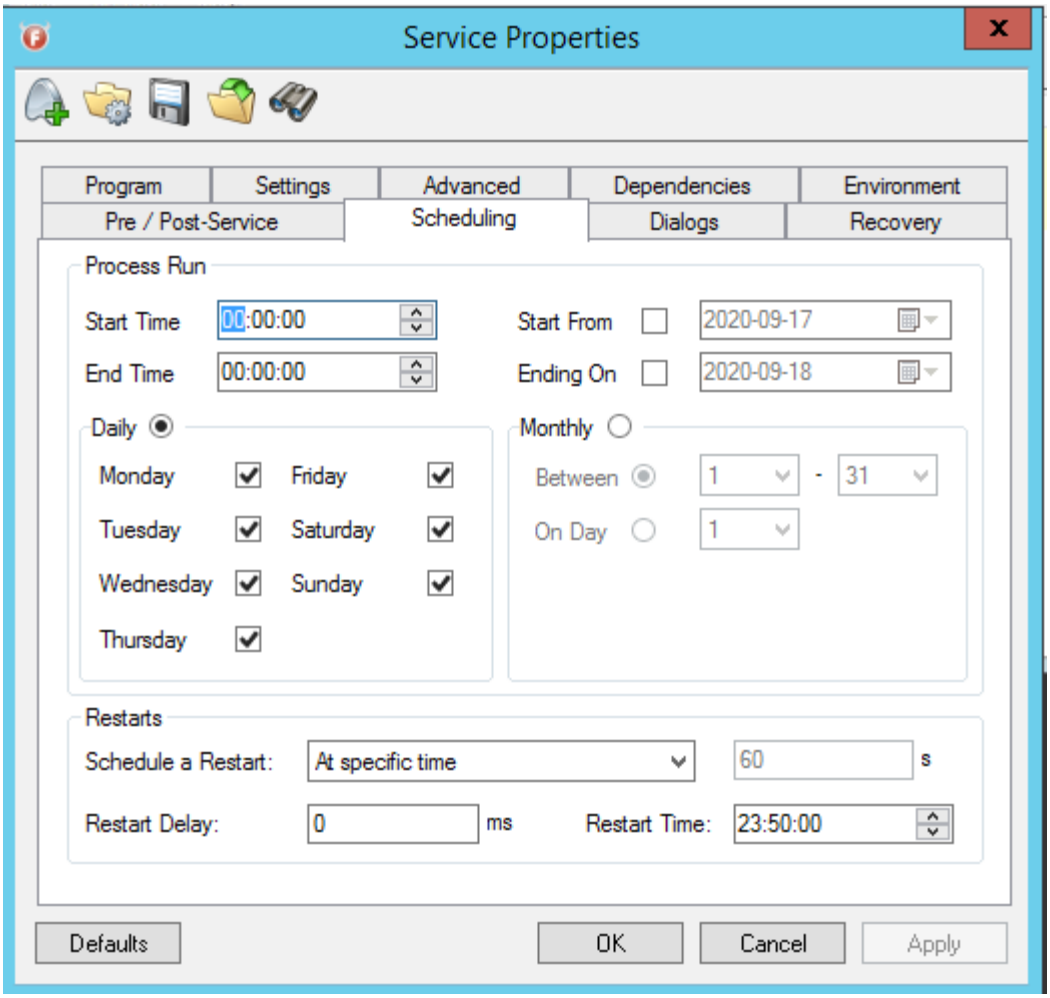
Pms service has specific start and end times. See screenshot below.



Key Service System Integration

FireDaemon properties implemented

Zip service has specific restart time. Process will then run once a day. See screenshot below.



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Winkey methods implemented

ControlKey by object – calculation of a access code. **TestAccess** by reference – alert error handling.

JSON sample from PMS (retrieving today's arrivals)

{"RetrievedAtUtc":"2018-12-23T09:04:37.0194133Z","Currency":"SEK","Bookings":
[{"BookingId":521481,"BookedBy":
{"Id":652016,"FirstName":"Keith","LastName":"Dawson","SocialSecurityNumber":"","StreetAddress":"Ringstrasse
19c","ZipCode":"5720","City":"Bremgarten","TelephoneNumber":"0041797276184","Email":"info@ettsmart.se","Co
mment":null,"IsObfuscated":false,"Name":"Keith
Dawson","CountryId":83,"OrganisationId":null,"HotelId":3389,"OrganisationName":null,"Country":"Schweiz","Memb
ershipNumber":null,"MarketingPermissions":false},"Rooms":
[{"BookingRowId":640925,"BillRowId":753369,"BookableObjectId":36547,"BookableObjectName":
{"Svenska":"StudioExtra 20","English":"StudioExtra 20"},"BookedFrom":"2018-12-
23T15:00:00","BookedTo":"2018-12-
27T11:00:00","BookingId":521481,"NumBeds":2,"NumNights":4,"CheckedIn":false,"CheckedOut":false}],"BillTotal"
:4780.00000000},{"BookingId":569752,"BookedBy":
{"Id":745317,"FirstName":"Stefan","LastName":"Holmberg","SocialSecurityNumber":"","StreetAddress":null,"ZipCo
de":"","City":"Ystad","TelephoneNumber":"+46722257307","Email":"info@ettsmart.se","Comment":null,"IsObfuscat
ed":false,"Name":"Stefan
Holmberg","CountryId":1,"OrganisationId":null,"HotelId":3389,"OrganisationName":null,"Country":"Sverige","Mem
bershipNumber":null,"MarketingPermissions":false},"Rooms":
[{"BookingRowId":702583,"BillRowId":827570,"BookableObjectId":36532,"BookableObjectName":
{"Svenska":"Dubbelrum 9","English":"Twin bed 9"},"BookedFrom":"2018-12-23T15:00:00","BookedTo":"2018-12-
26T11:00:00","BookingId":569752,"NumBeds":2,"NumNights":3,"CheckedIn":false,"CheckedOut":false}],"BillTotal"
:2385.00000000},{"BookingId":577928,"BookedBy":
{"Id":745310,"FirstName":"Pernilla","LastName":"Holmberg","SocialSecurityNumber":"","StreetAddress":"Ostvägen
44","ZipCode":"23252","City":"Åkarp","TelephoneNumber":"0724021352","Email":"info@ettsmart.se","Comment":n
ull,"IsObfuscated":false,"Name":"Pernilla
Holmberg","CountryId":1,"OrganisationId":null,"HotelId":3389,"OrganisationName":null,"Country":"Sverige","Mem
bershipNumber":null,"MarketingPermissions":false},"Rooms":
[{"BookingRowId":715280,"BillRowId":842492,"BookableObjectId":36533,"BookableObjectName":
{"Svenska":"Dubbelrum 10","English":"Twin bed 10"},"BookedFrom":"2018-12-23T15:00:00","BookedTo":"2018-
12-
25T11:00:00","BookingId":577928,"NumBeds":2,"NumNights":2,"CheckedIn":false,"CheckedOut":false}],"BillTotal"
:1590.00000000},{"BookingId":584910,"BookedBy":{"Id":591095,"FirstName":"TCS Sweden
Housing","LastName":"/Tata Consultancy Services Sverige AB","SocialSecurityNumber":"","StreetAddress":"Mäster
Samuelsgatan
42","ZipCode":"11257","City":"Stockholm","TelephoneNumber":"","Email":"sweden.arrival@tcs.com","Comment":n
ull,"IsObfuscated":false,"Name":"TCS Sweden Housing /Tata Consultancy Services Sverige
AB","CountryId":1,"OrganisationId":53225,"HotelId":3389,"OrganisationName":"TCS","Country":"Sverige","Member
shipNumber":null,"MarketingPermissions":false},"Rooms":
[{"BookingRowId":727006,"BillRowId":856045,"BookableObjectId":36552,"BookableObjectName":
{"Svenska":"Lägenhet 27","English":"Apartment 27"},"BookedFrom":"2018-12-21T15:00:00","BookedTo":"2019-
01-20T11:00:00","BookingId":584910,"NumBeds":1,"NumNights":30,"CheckedIn":true,"CheckedOut":false},
{"BookingRowId":727007,"BillRowId":856046,"BookableObjectId":36553,"BookableObjectName":
{"Svenska":"Lägenhet 28","English":"Apartment 28"},"BookedFrom":"2018-12-21T15:00:00","BookedTo":"2019-
01-20T11:00:00","BookingId":584910,"NumBeds":1,"NumNights":30,"CheckedIn":true,"CheckedOut":false},
{"BookingRowId":727008,"BillRowId":856047,"BookableObjectId":36557,"BookableObjectName":
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01-20T11:00:00","BookingId":584910,"NumBeds":1,"NumNights":30,"CheckedIn":true,"CheckedOut":false},
{"BookingRowId":727010,"BillRowId":856049,"BookableObjectId":36554,"BookableObjectName":
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01-20T11:00:00","BookingId":584910,"NumBeds":1,"NumNights":30,"CheckedIn":true,"CheckedOut":false},
{"BookingRowId":727011,"BillRowId":856050,"BookableObjectId":36555,"BookableObjectName":
{"Svenska":"Lägenhet 31","English":"Apartment 31"},"BookedFrom":"2018-12-21T15:00:00","BookedTo":"2019-
01-20T11:00:00","BookingId":584910,"NumBeds":1,"NumNights":30,"CheckedIn":true,"CheckedOut":false},
{"BookingRowId":727012,"BillRowId":856052,"BookableObjectId":36556,"BookableObjectName":
{"Svenska":"Lägenhet 32","English":"Apartment 32"},"BookedFrom":"2018-12-21T15:00:00","BookedTo":"2019-
01-
20T11:00:00","BookingId":584910,"NumBeds":1,"NumNights":30,"CheckedIn":true,"CheckedOut":false}],"BillTotal
":141120.0000000000000000}],"HotelTimeZone":"W. Europe Standard Time"}

Key Service System Integration

URL sample to PMS (update of access code)

https://api.happybooking.io/v1/DoorCodes?bookingRowId=695671&from=2018-09-22T17:00:00&to=2018-10-22T13:00:00&code=46*189715

XML sample to TellusTalk (email mailing)

```
<?xml version="1.0" encoding="UTF-8"?>
<root>
		<to>email:mary.jones@company.com</to>
		<email_from_name>SMART HOTEL</email_from_name>
		<reply_to>email: info@ettsmart.se</reply_to>
		<subject>Din Dörrkod</subject>
		<html>Dear/Kära gäst,

				Provided you will find your unique access code for your stay /din kod för vistelse.

				Reservation no/bokningsnr: 727429
				Name/Namn: Karlsson Jan
				Room/rum: 23
				Code/kod: Test # 1
				Arrival date/ankomst: 2020-02-17 , from/från 15.00
				Departure date/ avresa: 2020-02-19 , no later than/senast 11.00

				This message is also sent as a text message to the mobile number you provided at time of booking/Detta meddelande skickas även som text(SMS) till det mobilnr som uppgivits vid bokning.

				? : Send Mail	+46 8 350 260
		</html>
</root>
```

XML sample to TellusTalk (SMS mailing)

```
<?xml version="1.0" encoding="UTF-8"?>
<root>
		<to>sms:97593100</to>
		<subject>Din Dörrkod</subject>
		<text>Rum: 16
				Ank: 2018-05-24
				Avr: 2018-05-25
				Bokn nr: 517878
				vg. använd alla tecken &amp; siffror.Dörrkod: 16*617443
				Välkommen! För hjälp, ring oss på +468350260
		</text>
		<sms_originator_text>SMART HOTEL</sms_originator_text>
</root>
```